Digital Filter's Characteristics (Topics 2&3)



MATLAB can be used to determine important characteristics of a digital filter, including:

1. Transfer function
2. Impulse response
3. Frequency response (magnitude & phase)
4. Cut-off frequency, bandwitdth
5. Pole and zero pattern
6. Identify if the filter is:
   - FIR / IIR
   - Stable / unstable
   - Linear phase, min phase, max phase, mixed phase
   - All pass filter
   - Low pass, high pass, band pass, band stop

Digital Filter's Characteristics (Topics 2&3)

- Some MATLAB commands that are useful for filter analysis

MATLAB function	Purpose
roots	To find the zeros and poles of a filter from its transfer function
poly	To find coefficients of the characteristic polynomial from its roots.
zplane	To plot the pole-zero plane (zplane) of a filter from its poles, zeros or transfer function
freqz	To compute and/or plot the magnitude and phase response of a filter
filter	To compute the output of a filter from its transfer function and input
impz	To compute the impulse response of a filter from its transfer function
findpeaks	To finds the local peaks in the data vector Y
conv	To find the convolution of the two input vectors
fliplr	To flip an array in left/right direction

Hint: 'doc' or 'help' commands can be used to illustrate the usage of the above commands

```
>>doc filter
```

Example - Digital Filter Characteristics

Example 1: A discrete-time system described by its transfer function shown as below:

$$H(z) = \frac{z^4 - 0.4096}{z^3(z - 0.8)}$$

Using MATLAB,

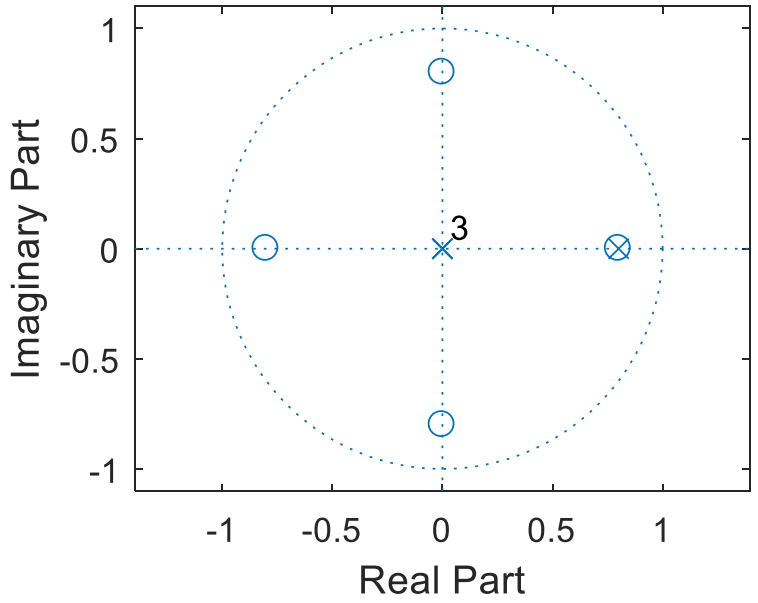
a. Find all the poles and zeros of the system and plot them in the z-plane
b. Compute and plot the magnitude response of the system
c. Compute and plot the impulse response of the system
d. Compute system's output y when the input signal is x = [1 1 1 2 2 2 3 3 3 4 4 4] using three different methods as given below:
    i. Using a for loop
    ii. 'conv' command
    iii. 'filter' command

Example - Digital Filter Characteristics

$$H(z) = \frac{z^4 - 0.4096}{z^3(z - 0.8)}$$

a. Find and plot poles & zeros

```
Zs = roots([1 0 0 0 -0.4096]); % roots of H(z)'s numerator polynomial
Ps= roots([1 -0.8 0 0 0]); % roots of H(z)'s denominator polynomial
zplane(Zs, Ps); % plot poles & zeros on z-plane
```



Example - Digital Filter Characteristics

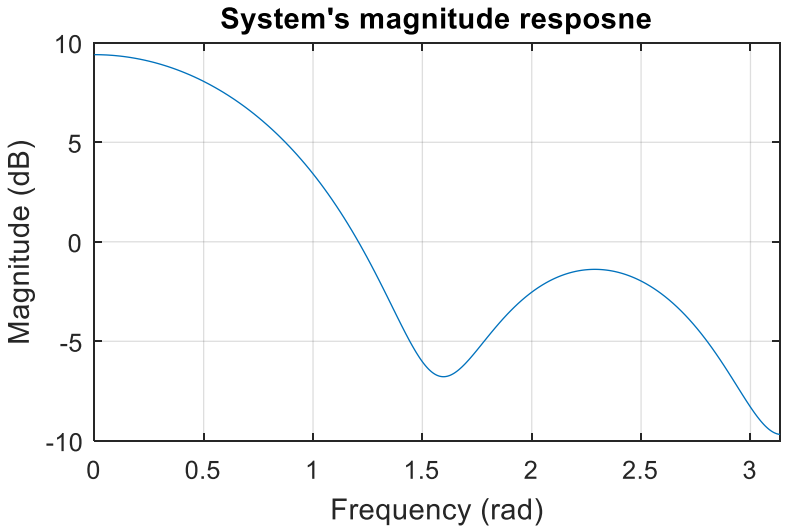
$$H(z) = \frac{z^4 - 0.4096}{z^3(z - 0.8)}$$

b. Compute and plot the magnitude response

H(z) can be re-written as

$$H(z) = \frac{z^{-4}(z^4 - 0.8^4)}{z^{-4}z^3(z - 0.8)} = \frac{(1 - 0.8^4 z^{-4})}{(1 - 0.8z^{-1})}$$

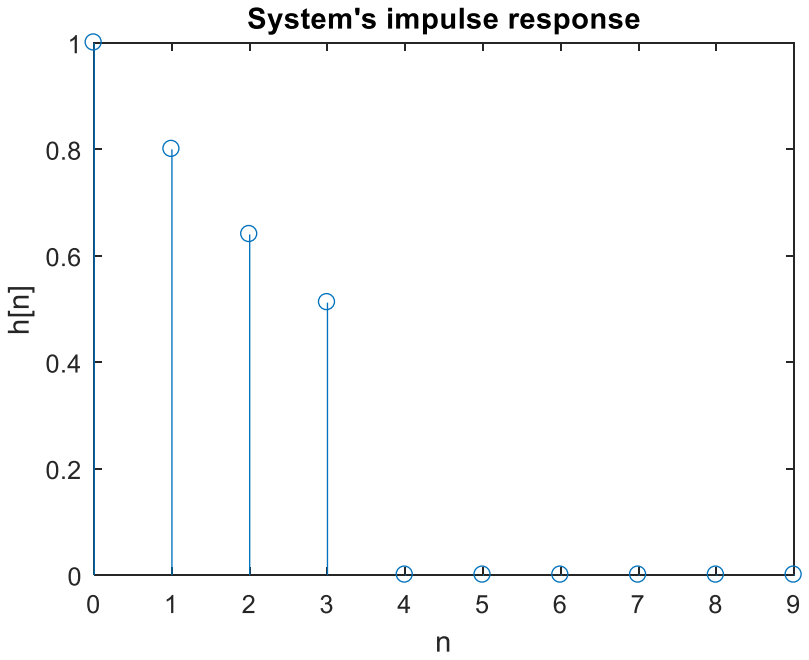
```
B=[1 0 0 0 -0.8^4];% coefficients. of H(z)'s numerator
A=[1 -0.8]; % coefficients of H(z)'s denominator
[H, w]=freqz(B, A);% compute system's frequency response
figure
plot(w,20*log10(abs(H)))% plot mag. response in dB scale
xlabel('Frequency (rad)');
ylabel('Magnitude (dB)')
title('System''s magnitude resposne')
xlim([w(1) w(end)]) % limit x-axis in range of w
grid on
```



Example - Digital Filter Characteristics

c. Compute and plot the impulse response of the system

```
[h, n] = impz(B, A);
stem(n, h)
xlim([n(1) n(10)])
xlabel('n');
ylabel('h[n]');
title('System''s impulse response')
```



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Example - Digital Filter Characteristics

d. Compute system's output y when the input signal is x = [1 1 1 2 2 2 3 3 3 4 4 4]

It can be seen from part (c) that only the first 4 elements of impulse response of the system are non-zero, i.e., $h[n] = [h_0, h_1, h_2, h_3]$

i. Using a for loop

```
x = [1 1 1 2 2 2 3 3 3 4 4 4];
h_nz = h(1:4); % non-zero elements of impulse response
x_extend = [x, zeros(1, length(h_nz)-1)]; % padding zeros to the end of x
y_fL = [];% output using for loop
y_fL(1)=h_nz(1)*x_extend(1);
y_fL(2)=h_nz(1)*x_extend(2)+h_nz(2)*x_extend(1);
y_fL(3)=h_nz(1)*x_extend(3)+h_nz(2)*x_extend(2)+h_nz(3)*x_extend(1);
h_nz_flip = fliplr(h_nz'); % flip impulse response
for i=4:length(x_extend);
    y_fL(i)=sum(h_nz_flip.*x_extend(i-4+1:i));
end
```

ii. Using MATLAB in-built command 'conv'

```
y_conv = conv(x, h_nz);
```


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Example - Digital Filter Characteristics

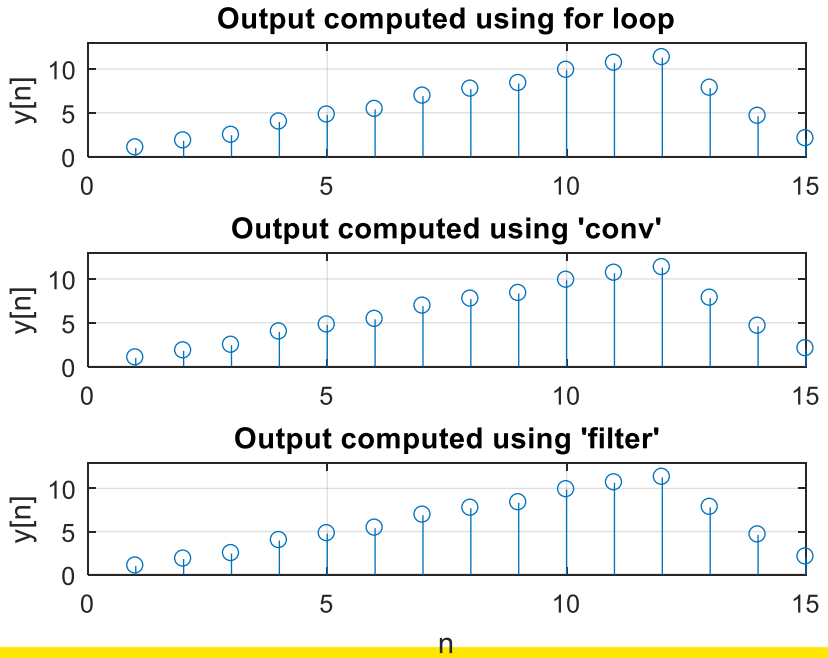
d. Compute system's output y when the input signal is x = [1 1 1 2 2 2 3 3 3 4 4 4]

It can be seen from part (c) that only the first 4 elements of impulse response of the system is non-zero, i.e., $h[n] = [h_0, h_1, h_2, h_3] = [1, 0.8, 0.64, 0.512]$

iii. Using MATLAB in-built command 'filter'

```
[y_filter, Zf] = filter(h_nz, 1, x);
 y_filter = [y_filter, Zf']; %concatenate final conditions to filter output
```

```
subplot 311
stem(y_fL)
ylim([0 13]); grid on; title('Output computed using for loop')
ylabel('y[n]')
subplot 312
stem(y_conv)
ylim([0 13]); grid on
title('Output computed using ''conv''''); ylabel('y[n]')
subplot 313
stem(y_filter)
ylim([0 13]); grid on
title('Output computed using ''filter''''); ylabel('y[n]'); xlabel('n')
```



Example – Frequency Estimation

Example 2:

a. Using MATLAB, generate 1024 samples of a discrete-time sinusoid signal with sampling rate fs=8000 and signal frequency at 450Hz.

b. Assume you are given the discrete-time signal in part (a) but you don't know the frequency of the sinusoid signal. Write a MATLAB code segment to estimate its frequency

Example – Frequency Estimation

Example 2:

a. Using MATLAB, generate 1024 samples of a discrete-time sinusoid signal with sampling rate fs=8000 and signal frequency at 450Hz.

```
N=1024;
n=0:N-1;
fs=8000;
fa=450;
x=sin(2*pi*fa*n/fs);
```

b. Assume you are given the discrete-time signal in part (a) but you don't know the frequency of the sinusoid signal. Write a MATLAB code segment to estimate its frequency

```
X=fft(x);
X=abs(X(1:N/2)); % take magnitude spectrum in frequency region [0, fs/2]
freqHz=(0:N-1)/N*fs; % Hertz frequency scale from 0 to fs
freqHz=freqHz(1:N/2); % Hertz frequency scale from 0 to fs/2
[pk, loc]=findpeaks(X);
estimated_signal_frequency = freqHz(loc)
```

Example – Frequency Estimation

Example 2:

```
subplot 211
plot(X)
xlim([0 length(X)-1]);
xlabel('frequency sample index')
ylabel('Magnitude')
title('Magnitude spectrum of signal')
grid on
subplot 212
plot(freqHz, X)
xlim([freqHz(1) freqHz(end)]);
xlabel('frequency (Hz)')
ylabel('Magnitude')
grid on
```

